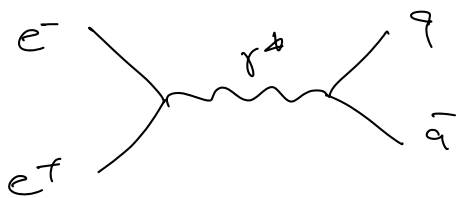


$$e^- e^+ \rightarrow \text{hadrons}$$



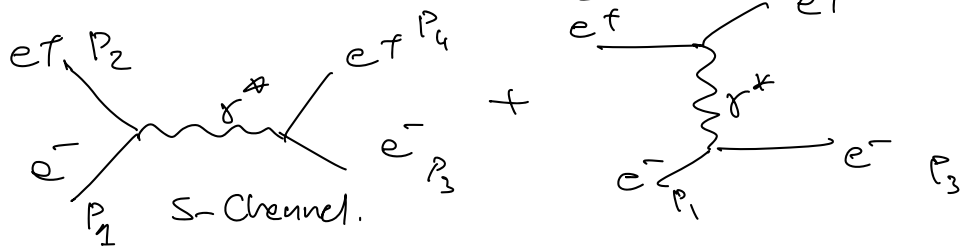
$$N_{\text{had}} = \sigma_{\text{had}} \cdot L_{\text{int}}$$

$$N_{e^+ e^-} = \sigma_{\text{Bhabha}} L_{\text{int}}$$

$$\Rightarrow \sigma_{\text{had}} = \frac{N_{\text{had}}}{N_{e^+ e^-}} \sigma_{\text{Bhabha}}$$

$$R = \frac{\sigma(e^+ e^- \rightarrow \text{hadrons})}{\sigma(e^+ e^- \rightarrow \mu^+ \mu^-)} \quad \text{why not } \frac{\sigma(e^+ e^- \rightarrow \text{had})}{\sigma(e^+ e^- \rightarrow e^+ e^-)}$$

$$e^+ e^- \rightarrow e^+ e^-$$



$$s = (\underline{p}_1 + \underline{p}_2)^2$$

$$t = (\underline{p}_1 - \underline{p}_3)^2 = -Q^2$$

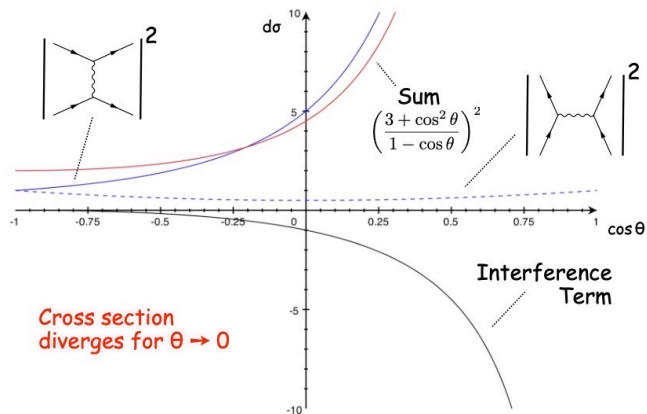
$$\sigma(e^+ e^- \rightarrow e^+ e^-) \propto |M_S + M_t|^2 = |M_S|^2 + |M_t|^2 + \underbrace{M_S^* M_t + M_t^* M_S}_{\text{interference}}$$

t-channel + s-channel

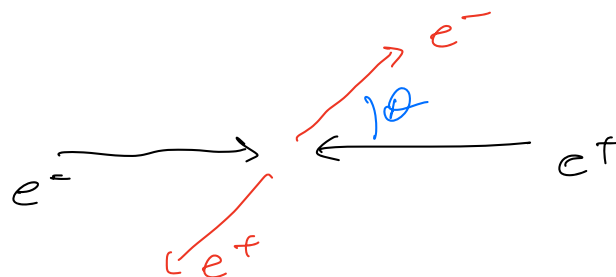
$$\frac{d\sigma_0}{d\Omega} = \frac{\alpha^2}{2s} \left(\frac{q'^4 + s^2}{q^4} + \frac{2q'^4}{q^2 s} + \frac{q'^4 + q^4}{s^2} \right)$$

interference term

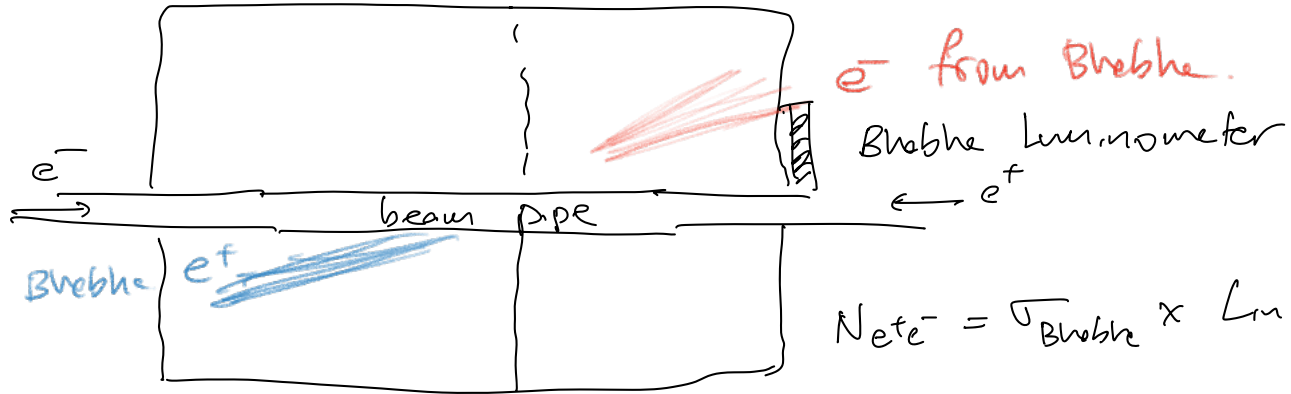
$$= \frac{\alpha^2}{4s} \left(\frac{3 + \cos^2 \theta}{1 - \cos \theta} \right)^2$$



Bhabha Scattering



In final state most electrons along direction of incoming e^-



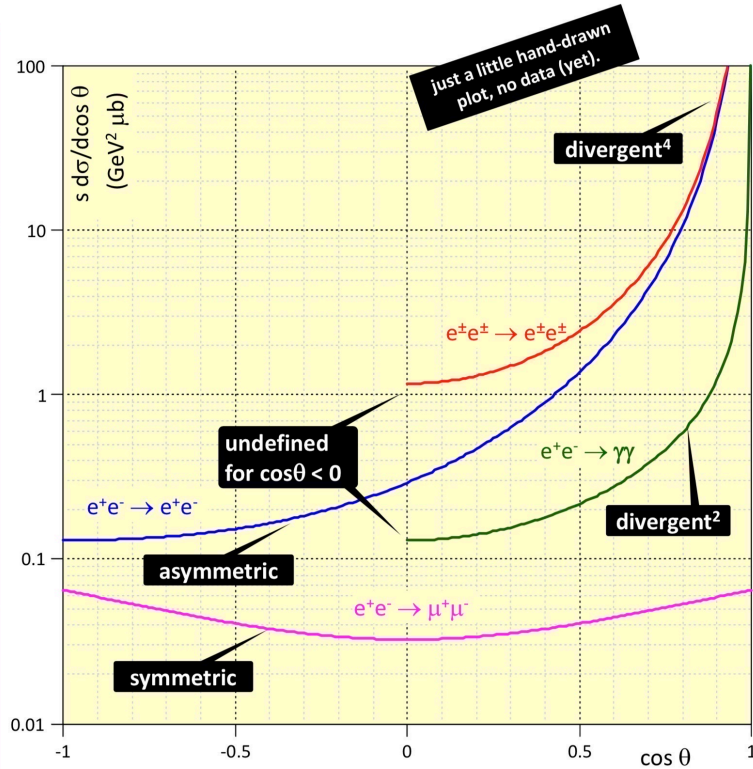
measure $L_{\text{int}} = \frac{N_e e^-}{\sigma_{\text{Bhabha}}}$

$$\frac{d\sigma(e^+e^- \rightarrow e^+e^-)}{d\cos\theta} = \frac{2\pi\alpha^2}{s} \times \left(\frac{3+\cos^2\theta}{1-\cos^2\theta} \right)^2;$$

$$\frac{d\sigma(e^+e^- \rightarrow \gamma\gamma)}{d\cos\theta} = \frac{2\pi\alpha^2}{s} \times \frac{1+\cos^2\theta}{1-\cos^2\theta};$$

$$\frac{d\sigma(e^+e^- \rightarrow e^+e^-)}{d\cos\theta} = \frac{\pi\alpha^2}{2s} \times \left(\frac{3+\cos^2\theta}{1-\cos^2\theta} \right)^2;$$

$$\frac{d\sigma(e^+e^- \rightarrow \mu^+\mu^-)}{d\cos\theta} = \frac{\pi\alpha^2}{2s} \times (1+\cos^2\theta);$$

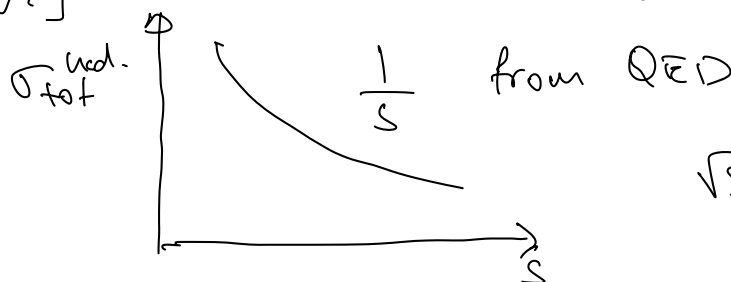


had.

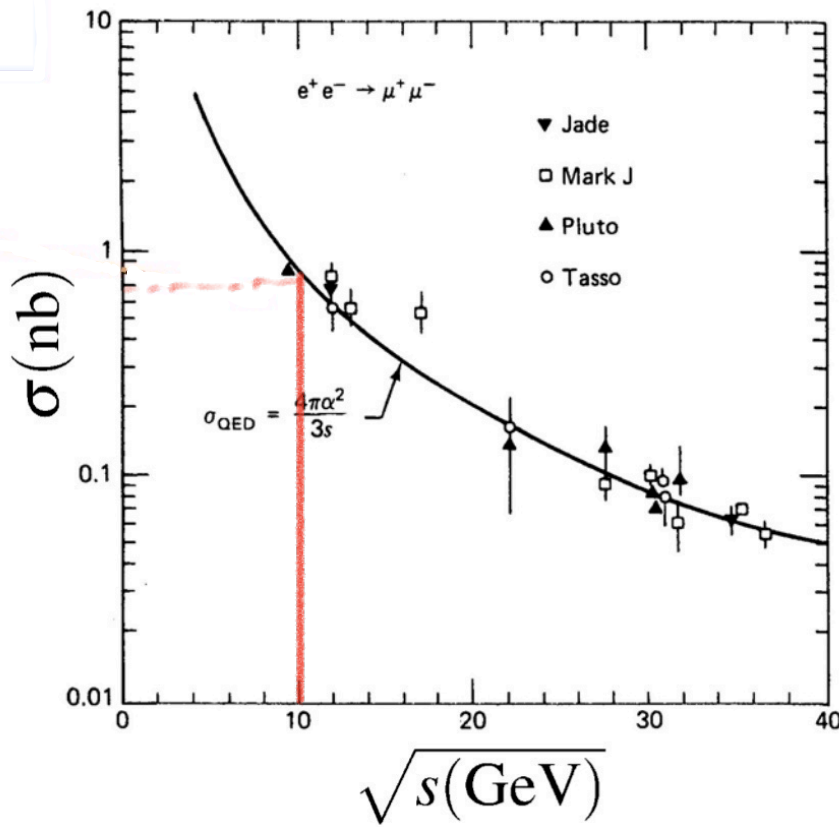
$$\sigma_{\text{tot}} = \frac{86.8 \text{ nb}}{s [\text{GeV}^2]}$$

$$\sigma_{\text{tot}}^{\text{had}} = \frac{N_{\text{had}}}{N_e e^-} \sigma_{\text{Bhabha}}$$

$\Rightarrow$

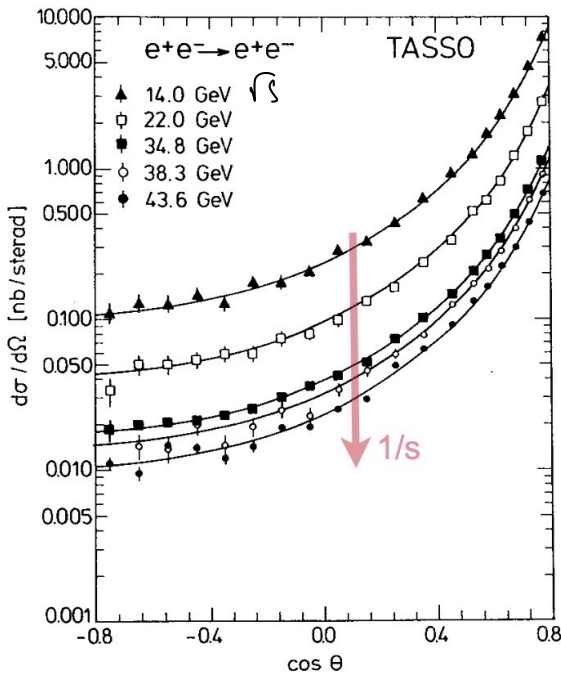


$$\sqrt{s} = 2 E_{\text{beam}}$$

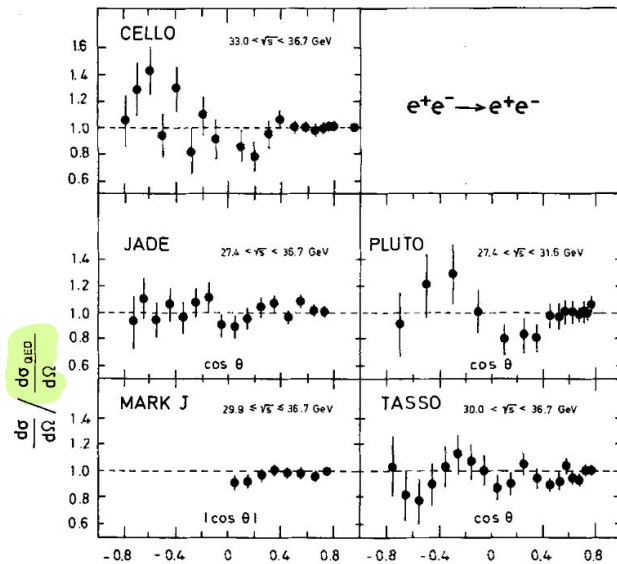


$\sqrt{s} = 10 \text{ GeV}$
 $\Rightarrow S = 100 \text{ GeV}^2$
 $\sigma = \frac{86.8 \text{ nb}}{100} \sim 0.9 \text{ nb}$
 total cross section
 as function of $\sqrt{s}$

$\frac{d\sigma}{d\Omega} \propto \frac{1}{s} (1 + \cos^2\theta)$
 predicted by QED



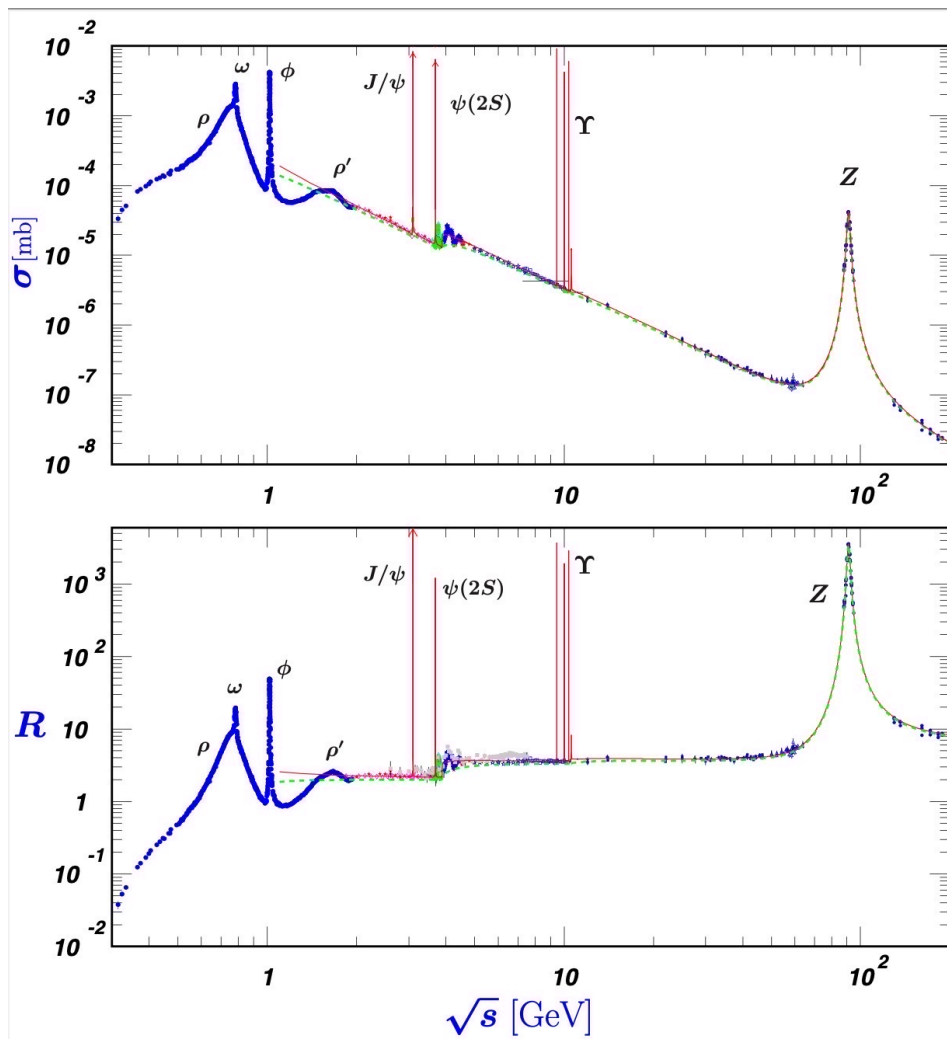
#events / theory prediction



Ratio between measured cross section and QED calculation

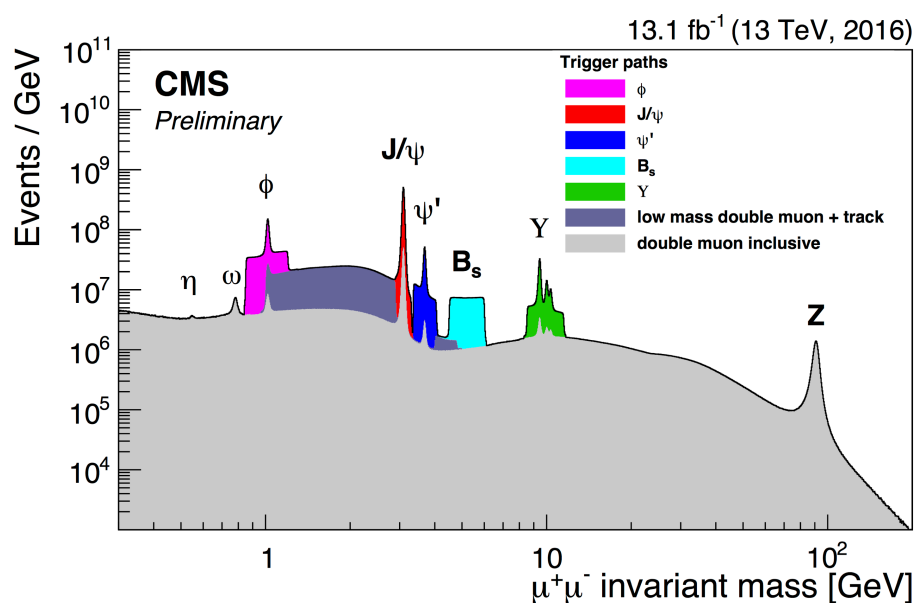
$\sqrt{s} = 14 \text{ GeV} \Rightarrow \sqrt{s'} = 43.6 \text{ GeV} \quad \sqrt{s'}/\sqrt{s} \sim 3$
 $\Rightarrow s'/s \sim 10 \Rightarrow \left. \frac{d\sigma}{d\Omega} \right|_{\sqrt{s}=14 \text{ GeV}} \sim 10 \times \left. \frac{d\sigma}{d\Omega} \right|_{\sqrt{s}=43 \text{ GeV}}$

QED predictions in good agreement w/ data for $e^+e^- \rightarrow \mu^+\mu^-$
 $\Rightarrow$ look at R for proof of flavor and color.

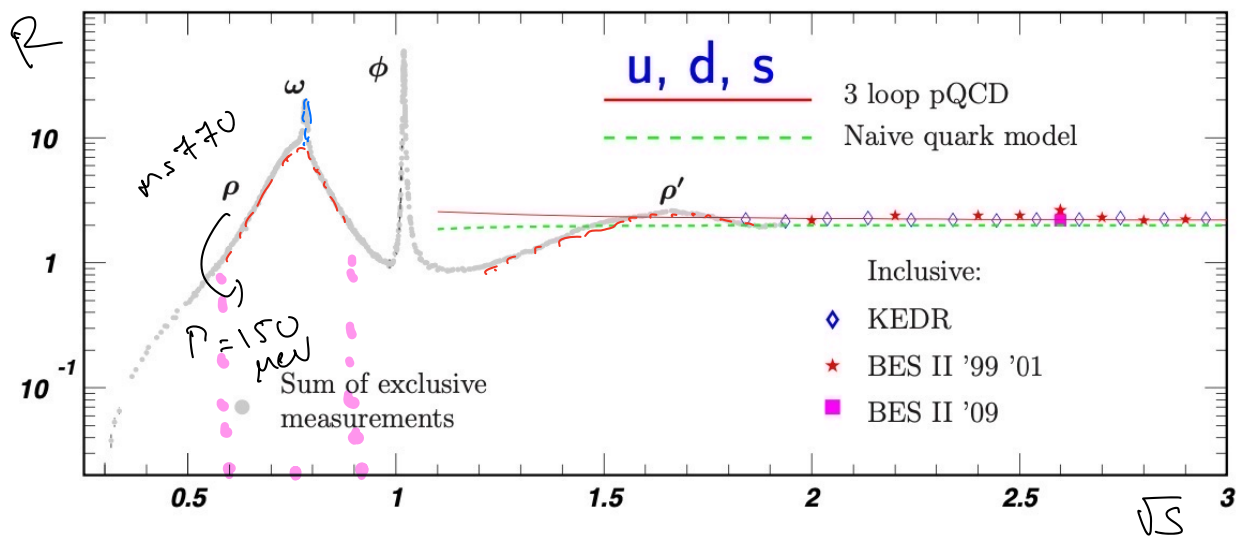


$$\sigma \sim \frac{1}{s}$$

$$R = N_c \times \sum_i Z_{q_i}^2$$



Same distribution produced in pp collisions @ LHC



$$\sqrt{s} < 3 \Rightarrow R = 3 \times \left(\left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 \right) = 3 \times \frac{4+1+1}{9} = \frac{6}{3} = 2$$

$\begin{matrix} u & d & s \\ \downarrow \\ N_c \end{matrix}$

$\Rightarrow$ in agreement with $N_c = 3$ hypothesis.
and 3 flavors

$\rho(770)$ $m = 770 \text{ MeV}$ $\omega(782)$ $m = 782 \text{ MeV}$
 vector meson $\Gamma = 150 \text{ MeV}$ $\Gamma = 8.7 \text{ MeV}$
 $\frac{u\bar{u} - d\bar{d}}{\sqrt{2}}$ ω also combination of $u\bar{u}, d\bar{d}$

$\phi \approx s\bar{s}$ $m = 1020 \text{ MeV}$ $\Gamma = 4 \text{ MeV}$

$\rho(770)$

$$I^G(J^{PC}) = 1^+(1^{--})$$

Isospin I

$$P = (-1)^{L+1} \quad f\bar{f}$$

$$C = (-1)^{L+S}$$

$$G = (-1)^{L+S+I}$$

$\omega(782)$

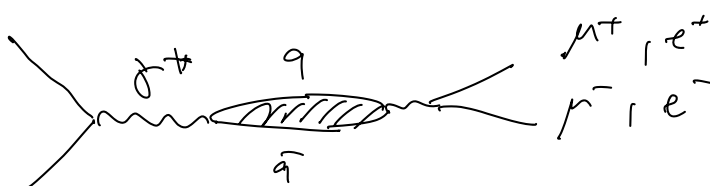
$$I^G(J^{PC}) = 0^-(1^{--})$$

$\phi(1020)$

$$I^G(J^{PC}) = 0^-(1^{--})$$

$$e^- : J = 1$$

No resonance with $J=0$



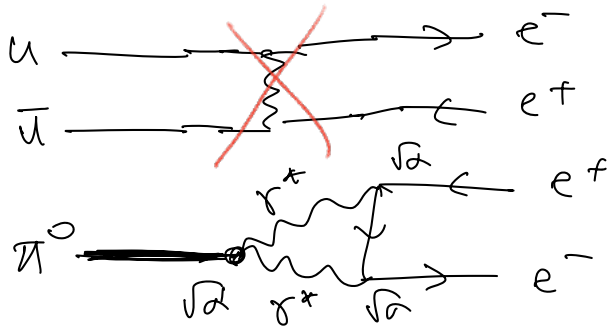
Resonance
must have same quantum numbers as photon.

$$J=1, P=-1, C=-1$$

why not π^0 ?

$$S = 0 \Rightarrow J = 0.$$

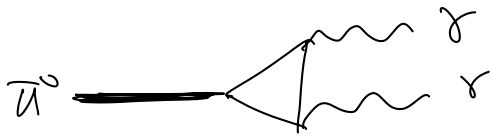
if $e^+e^- \rightarrow \pi^0 \Rightarrow \pi^0 \rightarrow e^+e^-$



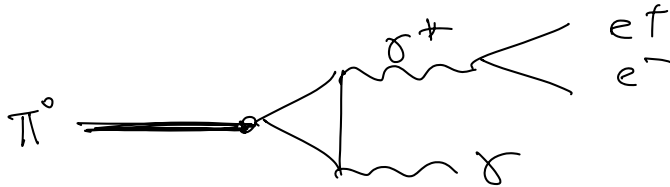
violates everything

2nd order diagram and is allowed.

$$M \propto \alpha^{3/2} \Rightarrow \sigma \propto |M|^2 \propto \alpha^3$$



$$\pi^0 \rightarrow \gamma\gamma$$

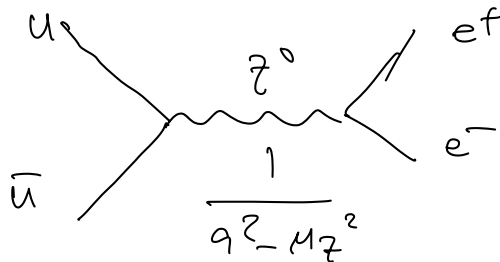


π^0 DECAY MODES

For decay limits to particles which are not established, see the appropriate Search sections (A^0 (axion) and Other Light Boson (X^0) Searches, etc.).

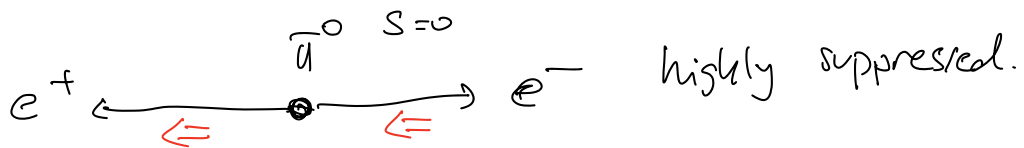
Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
Γ_1 2γ	(98.823 ± 0.034) %	S=1.5
Γ_2 $e^+e^-\gamma$	(1.174 ± 0.035) %	S=1.5
Γ_3 γ positronium	(1.82 ± 0.29) × 10 ⁻⁹	
Γ_4 $e^+e^+e^-e^-$	(3.34 ± 0.16) × 10 ⁻⁵	
Γ_5 e^+e^-	(6.46 ± 0.33) × 10 ⁻⁸	
Γ_6 4γ	< 2 × 10 ⁻⁸	CL=90%
Γ_7 invisible	< 4.4 × 10 ⁻⁹	CL=90%
Γ_8 $\nu_e\bar{\nu}_e$	< 1.7 × 10 ⁻⁶	CL=90%
Γ_9 $\nu_\mu\bar{\nu}_\mu$	< 1.6 × 10 ⁻⁶	CL=90%
Γ_{10} $\nu_\tau\bar{\nu}_\tau$	< 2.1 × 10 ⁻⁶	CL=90%
Γ_{11} $\gamma\nu\bar{\nu}$	< 1.9 × 10 ⁻⁷	CL=90%

$$\pi^0 = \frac{u\bar{u} - d\bar{d}}{\sqrt{2}}$$



$$q^2 \approx M_{\pi^0}^2$$

$$\mathcal{M}(\pi^0 \xrightarrow{\nu_{Z^0}} e^+e^-) \sim \frac{1}{M_Z^2} \text{ highly suppressed}$$



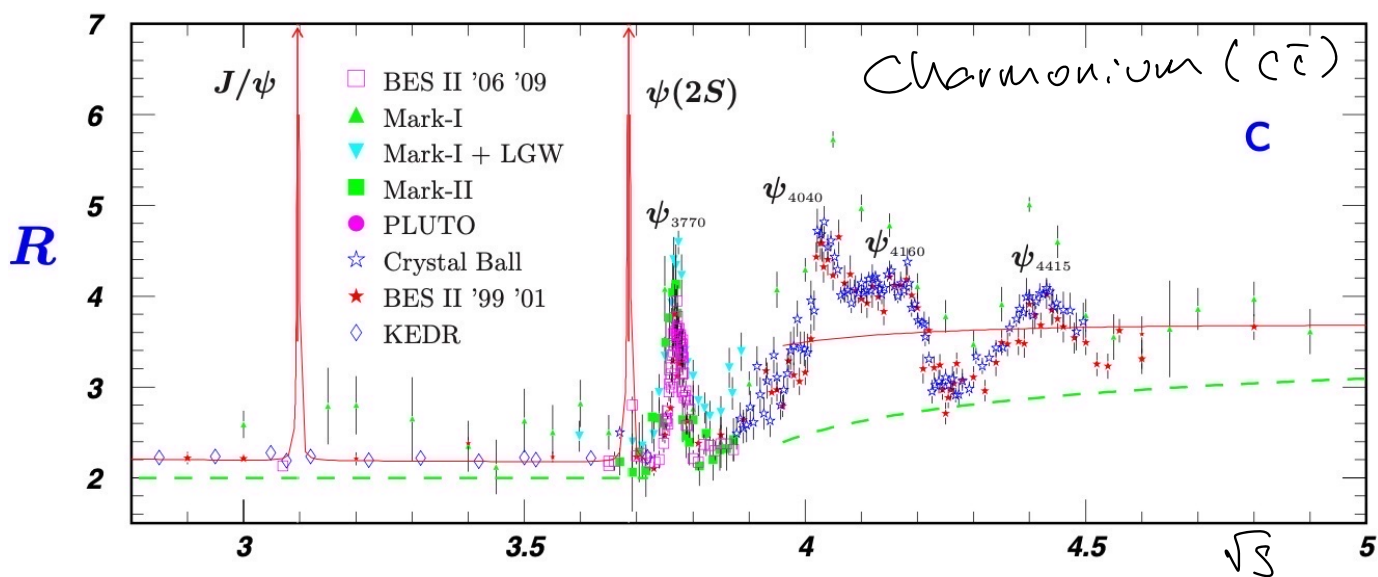
highly suppressed.

$$E_e = \frac{M_{\pi^0}}{2} = 70 \text{ MeV} \gg m_e = 0.5 \text{ MeV} \Rightarrow e^\pm \text{ are massless.}$$

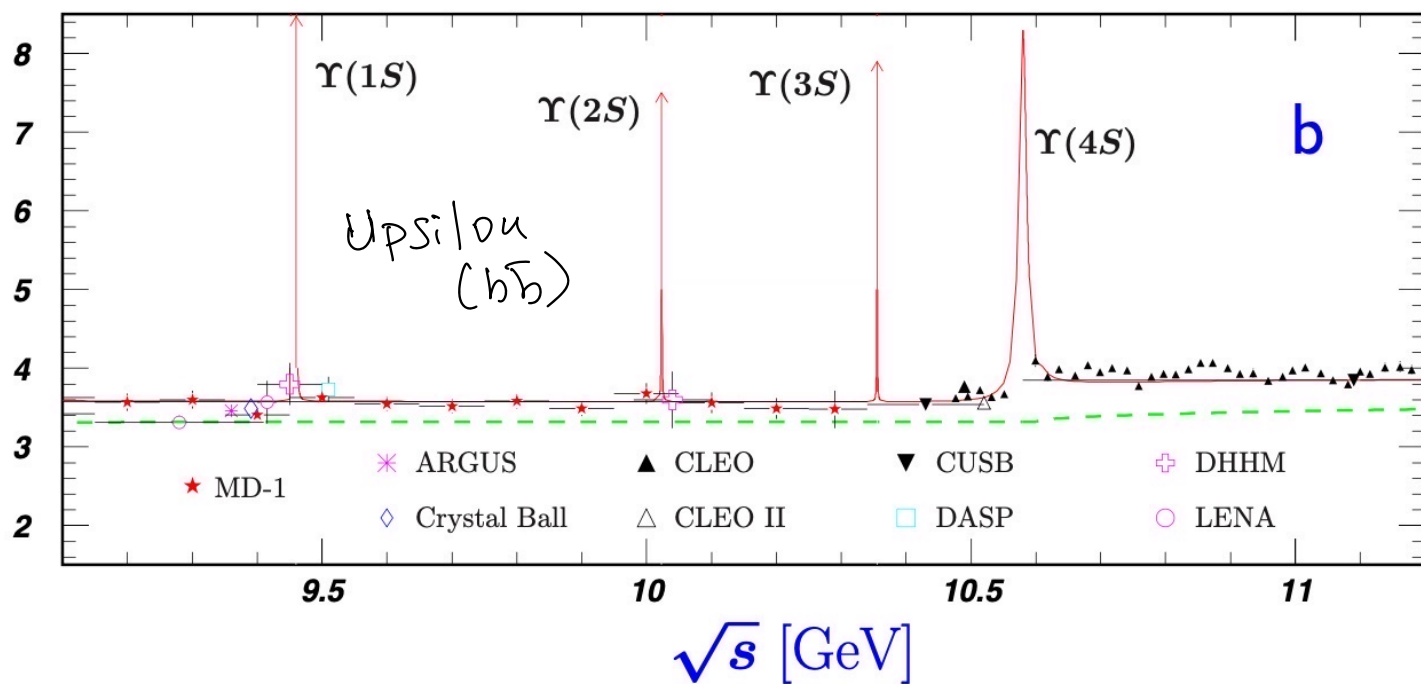
$$\beta\gamma = \frac{p}{m} = \frac{70}{0.5} = 140 \Rightarrow e^- \text{ mostly left handed}$$

$$J_Z^{\text{fin}} = \pm 1 \Rightarrow e^+ \text{ mostly right handed.}$$

Initial state is $S=0$ for π^0



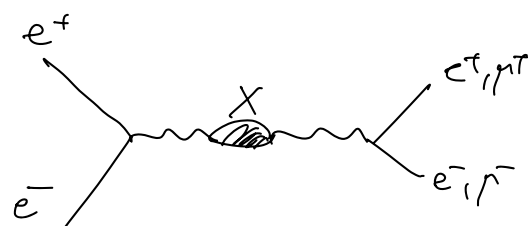
$\sqrt{s} \geq m_{J/\psi} \quad R = 2 + 3 \times \left(\frac{2}{3}\right)^2 = 2 + \frac{4}{3} = 3.33$



Discovery of new quarks

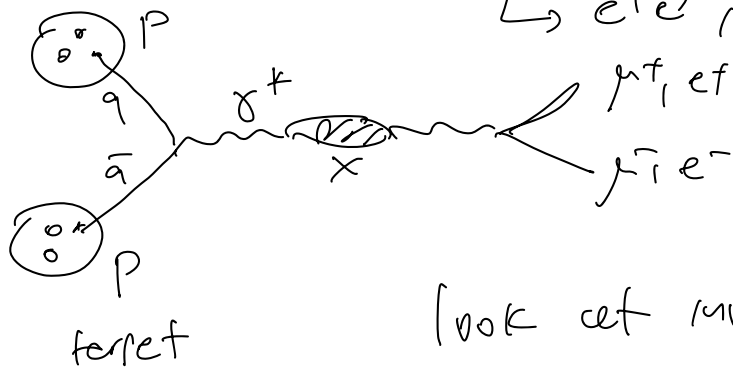
1) $e^+e^- \rightarrow X = \eta' \rightarrow \mu^+\mu^-, e^+e^-$

invariant mass $M_{\ell\ell^-}$ peak around m_X .



$$2) p + \text{target} \rightarrow X + X + \text{pions} + \dots$$

$$\hookrightarrow e^+e^-, \mu^+\mu^-$$

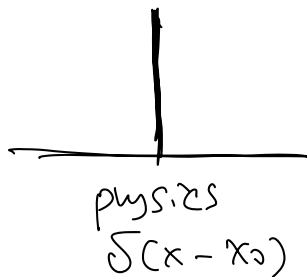


Drell-Yan process.
of production.

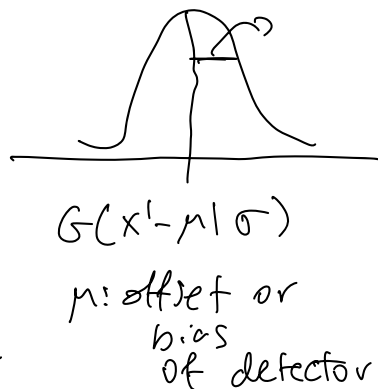
look at invariant mass.

$e^+e^-, \mu^+\mu^-$ better mass resolution than hadrons

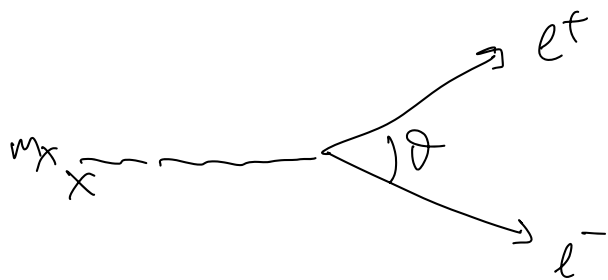
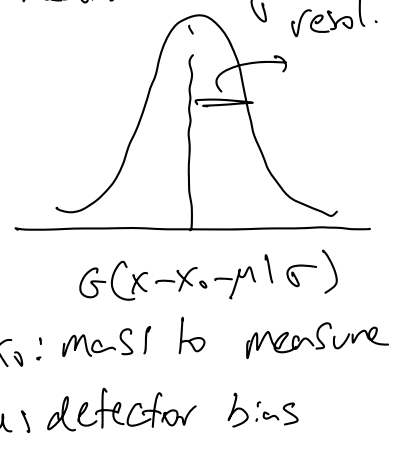
σ : resolution



$\otimes$



=



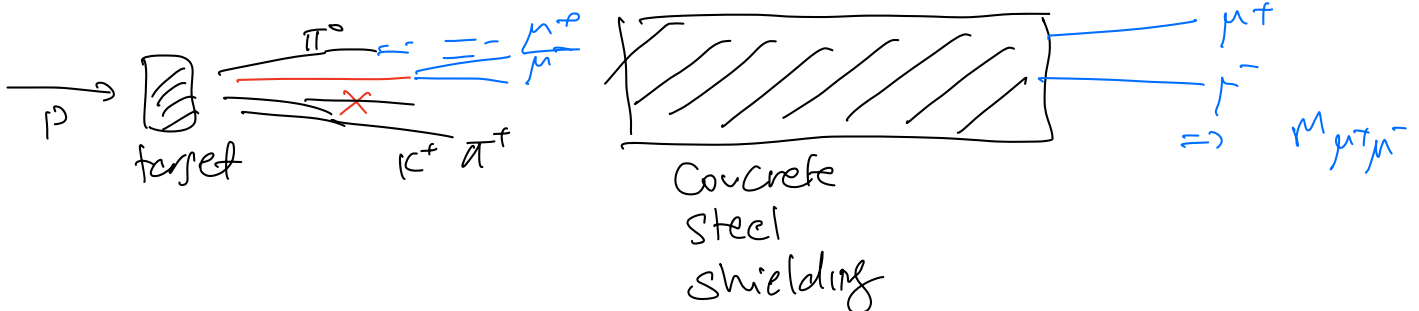
$$M_{\ell\ell}^2 = (E^+ + E^-)^2 - (\vec{p}^+ + \vec{p}^-)^2$$

$$\approx \sum E^+ E^- (1 - \cos\theta)$$

$$\approx 4 E^+ E^- \sin^2 \frac{\theta}{2}$$

Experimental resolution: $\sigma_m^2 = \sigma_E^2 \oplus \sigma_\theta^2$

$e^\pm$ have excellent resolution with EM calorimeters



Decay length: $\ell = \beta \gamma c \tau$

$p = 10 \text{ GeV}$

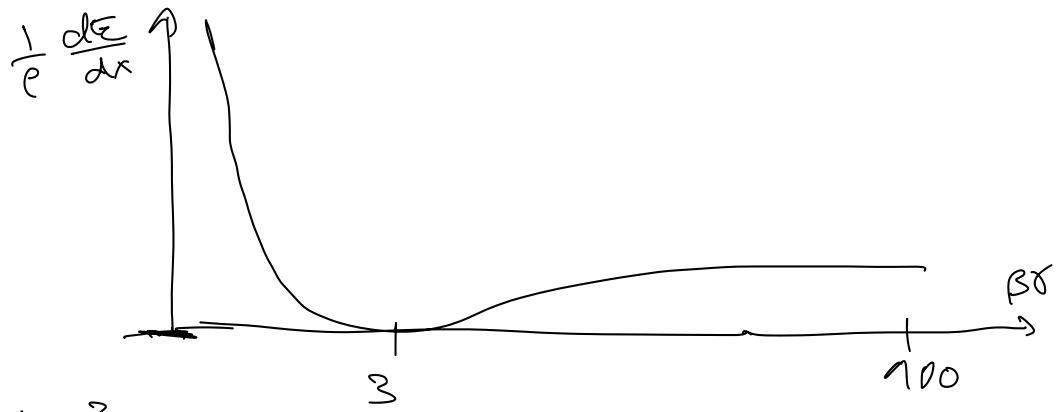
$\tau_\pi = 10^{-8}$

$\Rightarrow \ell = 10^2 \times 3 \times 10^{10} \frac{\text{cm}}{s} \times 10^{-8} s$

$\beta \gamma_\pi = \frac{10 \text{ GeV}}{0.14 \text{ GeV}} \approx 100$

$= 3 \times 10^4 \text{ cm} = 300 \text{ m}$

Bethe-Block:



$$MIP = 2 \frac{\text{MeV}}{\text{cm}} \frac{\text{cm}^3}{\text{g}}$$

assume $\rho = 10 \text{ g/cm}^3$

$$\Rightarrow \frac{dE}{dx} \approx 20 \text{ MeV/cm}$$

$$\Delta x = 3 \times 10^4 \text{ cm} \Rightarrow \Delta E = 3 \times 10^4 \times 20 \frac{\text{MeV}}{\text{cm}} \text{ cm}$$

$$= 6 \times 10^5 \text{ MeV} = \text{600 GeV}$$

over estimating

consider also Strong interaction

$\Rightarrow$ 100m of shielding to stop hadrons.

$\Rightarrow$ a lot of multiple Coulomb scattering for $\mu^+\mu^-$

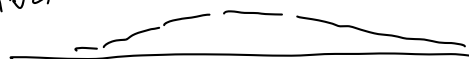
$$\sigma_m^2 = \sigma_E^2 \oplus \sigma_\theta^2$$

very bad after 100m of shielding.

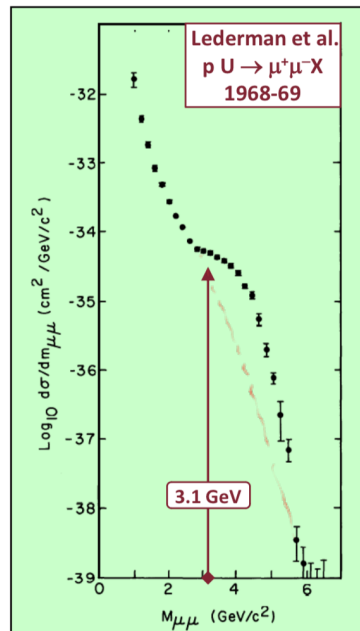


bad resolution

$\Rightarrow$



events



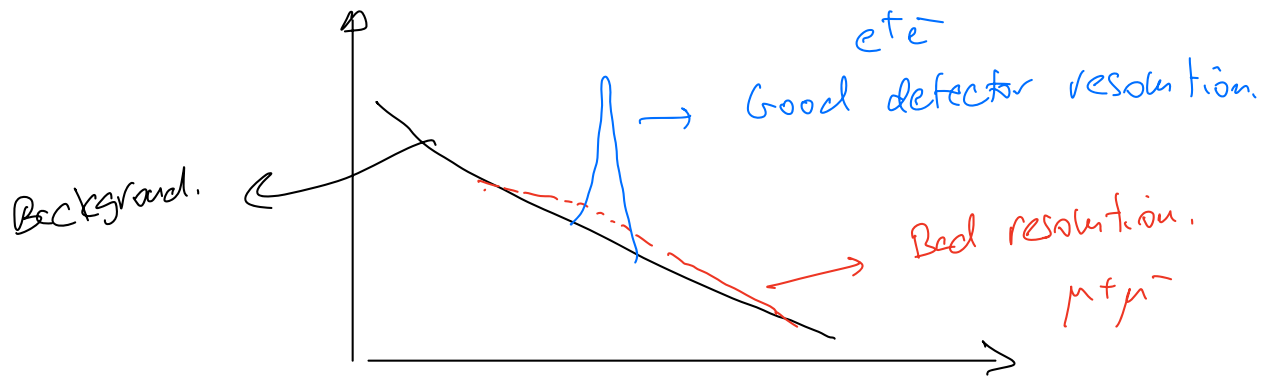
$\mu^+\mu^-$ invariant mass.

2 possibilities.

1) Threshold effect
Background



2) bad broad peak on top of background.



J/ψ discovery in 1974 (November Revolution)

1) SLAC: $e^+e^- \rightarrow c\bar{c} \rightarrow e^+e^-/\mu^+\mu^-/\text{hadrons}$.

2) Brookhaven: $p + \text{Be} \rightarrow X + e^- + e^- + e^-$
 $\hookrightarrow e^+e^-$

Look at the invariant mass e^+e^-